



Energy-efficient trajectory planning for UAV-assisted communication recovery using multi-agent graph reinforcement learning

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ABSTRACT

Unmanned aerial vehicle base stations (UAV-BSs) are effective and rapid to provide recovery of emergency communication after disasters due to their maneuverability. However, the throughput of mobile terminals (MTs) is prone to be limited by the trajectory and energy constraints of UAV-BSs. To improve throughput for MTs while guaranteeing energy efficiency of UAV-BSs, we propose an energy-efficient trajectory planning framework based on multi-agent heterogeneous graph reinforcement learning. We formulate the joint optimization problem as a partially observable Markov decision process. Then, we propose a heterogeneous graph-based method to represent relationships between UAV-BSs and network entities. Subsequently, we design a multi-agent graph attention recurrent actor-critic framework (MA-GAR) to efficiently learn over the heterogeneous graphs. Finally, we introduce a digital twin empowered centralized training and decentralized execution mechanism in MA-GAR to reduce energy consumption of UAV-BSs. Experimental results show that the proposed MA-GAR outperforms the benchmark algorithms in convergence speed, system throughput, energy consumption, and service fairness.

1. Introduction

With the advancement of wireless communication technologies, 5G networks have been widely deployed, and the next-generation network (6G) is poised to emerge [1]. These networks must achieve seamless connectivity, elevated communication rates, and green communication [2]. Unmanned aerial vehicles (UAVs), with substantial flexibility and rapid deployment capabilities, are widely used in Internet of Things (IoT) scenarios such as disaster relief [3], data collection [4], and task offloading [5].

Deploying UAVs as aerial base stations in wireless communication networks is a noteworthy advancement [6]. Especially in disaster areas with communication resource shortages, UAV-BSs can be temporarily deployed in the air for rapid assistance [7]. They can adjust their flight trajectories to provide wide-area, low-cost wireless communication coverage for ground terminals when ground facilities are damaged or severely congested [8].

Although UAV-BSs have various advantages in the communication fields, they still face challenges. Due to the limitation of onboard battery capacity of UAVs, their service time is limited, and great attention should be given to the energy efficiency [9,10]. Additionally, deep reinforcement learning methods have been demonstrated to have

excellent performance in controlling UAVs to improve the performance of wireless communication networks. Multi-agent deep reinforcement learning (MADRL) allows each UAV-BS to become a decision-maker who continuously optimizes its strategy by learning from training data generated through the environment. However, due to the limited computing resources of UAV-BSs, the traditional decentralized independent training mechanism is impractical. Because this mechanism requires extra energy consumption to achieve an optimal strategy, it can further reduce energy efficiency. Therefore, it is essential to design a more efficient and reasonable trajectory planning mechanism.

Most existing works on UAV-assisted communication service mainly focus on serving static ground terminals [11], which do not require frequent position adjustments of UAV-BSs. However, mobile terminals (MTs) require UAV-BSs to adjust their trajectories in real time to guarantee high-quality and stable communication services for their movements. The movement of UAV-BSs and MTs can alter the network topology and result in intermittent connections or even transmission failures [12]. These problems can be alleviated by optimizing the three-dimensional (3D) spatial layout of UAV-BSs, which is definitely an extremely complex problem [13]. The flight altitude of UAV-BS

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directly determines their coverage area. Different geographical environments and MT distributions require different flight altitudes to optimize the coverage effect.

In the existing UAV-BSs trajectory planning research, the locations of all ground terminals are assumed to be known in advance, which is usually unrealistic, especially in disaster relief scenarios [14]. Due to the attenuation of wireless transmission signals, the sensing range of UAV-BSs is limited, allowing UAV-BSs to observe only local MTs or other UAV-BSs. It is impossible to acquire the exact locations of all MTs in real time. Therefore, it is difficult to deploy them accurately according to the requirements of communication coverage. Thus, UAV-BSs must accurately plan their flight routes using local observations in practice. This requires them to fully utilize local data, autonomously search for MTs based on experience, and provide high-quality, stable communication link coverage with minimal energy consumption. Additionally, the movement of both UAV-BSs and MTs makes the network highly dynamic. Moreover, the large-sized partially observed information with time variations leads to inconsistent input dimensions. These problems pose insurmountable challenges for traditional convex optimization techniques and MADRL [9].

Considering the high dynamicity, partial observation constraints, and energy consumption limitations of UAV-BSs, we focus our research on energy-efficient trajectory planning. We thus designed a multi-agent graph attention recurrent actor-critic framework (MA-GAR) and introduced graph neural networks (GNNs) [15,16]. We use the graph structure to model the relationships between network entities. We use GNNs to capture their spatial dependencies in order to address the fluctuating dimension problem caused by dynamic and partially observable environments. Given the limited onboard computing power and energy of UAV-BSs, we introduced a centralized training and decentralized execution (CTDE) mechanism [17,18] into MA-GAR. This mechanism enables UAV-BSs to realize remote centralized training and decentralized parameter distribution, which can be used to improve the energy efficiency of UAV-BSs. However, UAV-BSs must continuously communicate with the central server to acquire state information in a centralized training phase of CTDE, which will incur high communication costs [19]. To mitigate this issue, we introduce digital twins (DT) into MA-GAR. DT technology maintains high-fidelity, real-time virtual representations of the physical world, enabling continuous monitoring of actual network states [20]. As an extensively adopted industrial solution, DT leverages its powerful computational resources and global perspective to provide UAV agents with a highly realistic and efficient training simulation environment. By integrating a DT-empowered CTDE mechanism into MA-GAR, we effectively reduce communication energy consumption. The main contributions are summarized as follows:

1. We propose a dynamic and uncertain air-ground integrated network with partial observability constraints and formulate a joint optimization problem for maximizing MTs' throughput and UAV-BSs' energy efficiency.
2. We formulate the problem as a partially observable Markov decision process (POMDP) and use heterogeneous graphs to represent the relationships between UAV-BSs and network entities to better describe the environment.
3. We design MA-GAR with a multi-agent graph attention recurrent actor-critic structure to flexibly handle local observations of fluctuating dimensions and efficiently learn over heterogeneous graphs.
4. We introduce a DT-empowered CTDE mechanism in MA-GAR, in which agents can more easily collect a large amount of timely interaction data to perform more efficient centralized training and further reduce energy costs.

The remainder of this paper is organized as follows: Section 2 reviews related literature. Section 3 describes the system model and problem formulation. Section 4 introduces the solution. The numerical results are analyzed in Section 5. Section 6 summarizes the paper and discusses future research directions.

2. Related works

This paper outlines related research work in the field of communication network optimization from Section 2.2 to 2.4. In Section 2.5, by comparing with existing studies, the unique advantages of the proposed MA-GAR framework are elucidated.

2.1. Traditional convex optimization algorithms

Wu and Zhang [21] assumed that the positions of ground users were known and static, and jointly optimized the resource allocation strategy and the UAV's trajectory to maximize throughput. They proposed a block coordinate descent (BCD) approach to solve the problem. Wang et al. [22] assumed static ground users and a single UAV-BS serving them. They adopted successive convex approximation (SCA) with Dinkelbach's algorithm to formulate a convex problem, which was solved using the BCD approach. In these works, convex optimization techniques were introduced to solve the optimization problems under ideal conditions.

2.2. MADRL-based optimization algorithms

Recent literature has focused on MADRL-based approaches. MADRL can effectively optimize system efficiency in complex communication scenarios. Zhao et al. [23] proposed a distributed MADRL method based on deep Q-network (DQN) to achieve maximum overall network utility from a long-term perspective. Arain et al. [24] adopted the multi-agent deep deterministic policy gradient (MADDPG) algorithm to achieve low-energy and low-latency scheduling strategies in heterogeneous wireless networks. Song et al. [25] proposed a centralized MADRL multi-objective joint optimization method for emergency scenarios, which optimizes system latency and energy consumption through double deep q-network (DDQN) and global parameter aggregation. The above studies fully demonstrate the effectiveness of MADRL in optimizing performance in fully observable communication scenarios. However, real-world communication environments are often more challenging. Due to transmitted signal attenuation, the observation range of network entities is limited. Moreover, entity mobility generates substantial data of fluctuating dimensions, making it difficult for DRL methods to learn efficiently from such samples Zhang et al. [26].

2.3. Optimization algorithms integrating MADRL with DT

Due to its powerful abstract description and analysis capabilities, recent studies have combined DT with UAV-assisted related fields to improve efficiency in strategy planning. Li et al. [27] combined the DT and MADRL model with a UAV-assisted perception, communication, and computing network to provide intelligent offloading decisions in DT. Liu et al. [28] constructed a UAV-assisted mobile edge computing scenario to enhance services for mobile users by incorporating DT. Wang et al. [29] investigated computation offloading in a UAV-assisted mobile emergency network based on DT disaster scenarios, leveraging DT's real-time prediction capability to optimize computation offloading strategies. Tanaka et al. [30] proposed a post-disaster communication recovery solution for UAVs that integrates DT technology, optimizing deployment strategies by constructing virtual replicas of the disaster-affected areas.

2.4. Optimization algorithms combining GNNs with MADRL

Recently, the innovative integration of GNNs and MADRL in dynamic communication scenarios has garnered widespread attention from researchers. Xie et al. [31] proposed a method combining Graph convolutional networks (GCNs) with proximal policy optimization (PPO) for fully observable device-to-device (D2D) networks. This approach utilizes GCN to learn dependency relationships within D2D

Table 1
Multi-dimensional comparison between existing works and MA-GAR.

Reference	UAV freedom	Terminal	Observability	Method	Objective	Train&Execute
[21]	Fixed altitude	Static	Fully	Convex optimization	Throughput	Not specified
[22]	Fixed altitude	Static	Fully	Convex optimization	Energy	Not specified
[33]	Fixed altitude	Dynamic	Fully	DRL	Throughput, Energy	Not specified
[25]	Fixed altitude	Static	Fully	MADRL	Latency, Energy	Not specified
[34]	Fixed altitude	Static	Fully	MADRL	AoI, Energy	Not specified
[6]	Fixed altitude	Static	Fully	MADRL	Throughput, Energy	Not specified
[18]	Fixed altitude	Dynamic	Partially	MADRL	Latency, Energy	CTDE
[35]	Fixed altitude	Static	Partially	MADRL	Throughput, Fairness	Not specified
[26]	Fixed altitude	Dynamic	Partially	GNNs, MADRL	Throughput, Fairness	CTDE
[31]	Fixed altitude	Dynamic	Fully	GNNs, MADRL	Throughput	CTDE
MA-GAR	Variable altitude	Dynamic	Partially	GNNs, RNN, MADRL	Throughput, Energy	DT, CTDE

networks and employs PPO to optimize the system's quality of experience (QoE). Sun et al. [32] introduced a task offloading framework that integrates GNNs with the actor-critic architecture for partially observable multi-access mobile edge computing (MEC) scenarios. This framework models the MEC system as an acyclic graph and uses GNNs to learn offloading strategies for each agent.

The aforementioned studies investigate environments with different observation capabilities, where the degree of observability significantly impacts the learning efficiency of GNNs. In partially observable environments, GNNs can only learn features from local subgraphs, making it difficult to capture global information. Furthermore, GCN typically employs simple message aggregation algorithms, which limit learning efficiency. To address these limitations, our proposed MA-GAR framework incorporates a multi-head attention mechanism and a recurrent mechanism. The former assigns varying weights to information from MTs, while the latter enables the indirect inference of global information. Through their synergistic effect, these components significantly enhance the learning efficiency of MA-GAR, enabling it to approach the optimum solution to the problem.

2.5. Summary

Table 1 compares MA-GAR with existing works across six dimensions: flight freedom, terminal mobility, environment observability, optimization method, objective, and training & execution method. Different from the works above, we consider a dynamic and partially observable communication network scenario. In this scenario, UAV-BSs fly at variable altitudes, offering greater freedom of movement and a wider explorable space. We propose the MA-GAR framework and incorporate GNNs, an attention mechanism, and a recurrent mechanism to efficiently learn energy-efficient trajectory planning strategies in challenging environments. In addition, we introduce a DT-empowered CTDE mechanism to further reduce energy costs and ensure real-time performance, thereby better meeting practical application scenarios.

3. System model and problem formulation

This section first introduces the system model, then presents the optimization problem. Table 2 summarizes the key notations used in this paper.

3.1. System model

Fig. 1 shows an air-ground integrated network consisting of UAV-BSs, MTs, and an edge computing center (ECC). There is a $L(m) \times L(m)$ ground area without ground base stations. There are M MTs with random movement, denoted by set $\mathbb{M} \triangleq \{1, 2, \dots, M\}$. U UAV-BSs denoted by the set $\mathbb{U} \triangleq \{1, 2, \dots, U\}$ flying above the area with limited power supply. A digital twin edge network (DITEN) [36] is deployed in the ECC. The ECC includes an edge computing server (ECS), which is continuously powered for data storage and model training, and a ground base station (BS) for state synchronization. We evenly divide the service period into time slots as $t \in \{1, 2, \dots, T\}$, the duration of each time slot is Δt .

3.1.1. Flight model of UAV-BSs

The flight altitude of UAV u at time slot t is denoted by $h_u^{\text{UAV}}(t)$. To fully utilize the flexibility of UAV-BSs, we do not fix the flight altitude, but limit the flight range to $h_u^{\text{UAV}} \min \leq h_u^{\text{UAV}}(t) \leq h_u^{\text{UAV}} \max$. It is equipped with a directional antenna that transmits energy to the ground at a fixed angle θ_{ts} . The ground coverage radius at time slot t of UAV u is denoted by $R_u^{\text{Cov}}(t) = \tan(\theta_{ts}) \cdot h_u^{\text{UAV}}(t)$, where $R_u^{\text{Cov}}(t)$ increases as the altitude $h_u^{\text{UAV}}(t)$ increases. The position of UAV u at time slot t is denoted by $\mathbf{p}_u^{\text{UAV}}(t) = [x_u^{\text{UAV}}(t), y_u^{\text{UAV}}(t), z_u^{\text{UAV}}(t)]^T$. To prevent collisions, all UAV-BSs are limited to flying outside the minimum safe distance $d_{\min}(m)$. The flight direction of UAV u is $\theta_u^{\text{UAV}}(t) = (\theta_u^{\text{UAV,H}}(t), \theta_u^{\text{UAV,V}}(t))$, where $\theta_u^{\text{UAV,H}}(t)$ denotes the flight angle in the horizontal plane and $\theta_u^{\text{UAV,V}}(t)$ denotes the flight angle in the vertical plane. Each UAV-BS is stationary at the beginning of a time slot, then chooses a fly action and accelerates $a_u^{\text{UAV}}(t)(m/s^2)$ for a duration of $\frac{\Delta t}{2}(s)$, and subsequently decelerates $a_u^{\text{UAV}}(t)$ for the same duration until it becomes stationary again. The instantaneous velocity of UAV u during movement is denoted by $\mathbf{v}_u^{\text{UAV}}(t)$. Thus, the position of UAV u can be updated by

$$\mathbf{p}_u^{\text{UAV}}(t+1) = \begin{bmatrix} \frac{4x_u^{\text{UAV}}(t) + a_u^{\text{UAV}}(t)\Delta t^2 \cos(\theta_u^{\text{UAV,H}}(t))}{4} \\ \frac{4y_u^{\text{UAV}}(t) + a_u^{\text{UAV}}(t)\Delta t^2 \sin(\theta_u^{\text{UAV,H}}(t))}{4} \\ \frac{4z_u^{\text{UAV}}(t) + a_u^{\text{UAV}}(t)\Delta t^2 \sin(\theta_u^{\text{UAV,V}}(t))}{4} \end{bmatrix}. \quad (1)$$

3.1.2. Sensing model of UAV-BSs

UAV-BSs can sense the state information of other nearby UAV-BSs and MTs through a non-payload communication (CNPC) link [37]. However, due to signal attenuation, the sensing distance of UAV-BSs is limited. Let $d_{\text{sns}}(m)$ denote the sensing distance of UAV-BSs m . The radius of the ground that UAV u can sense at time slot t is denoted by $R_u^{\text{Sns}}(t) = \sqrt{d_{\text{sns}}^2 - h_u^{\text{UAV}}(t)^2}$, where $R_u^{\text{Sns}}(t)$ decreases as the altitude $h_u^{\text{UAV}}(t)$ increases. The sensing matrix $\Omega^{\text{Sns}}(t) \in \{0, 1\}^{U \times M}$ denotes the MT sensing situation of all UAV-BSs. $\Omega_{u,m}^{\text{Sns}}(t) = 1$ means UAV u can sense the state information of MT m , and 0 if not. The sensing matrix $\Lambda^{\text{Sns}}(t) \in \{0, 1\}^{U \times (U-1)}$ denotes the sensing situation between all UAV-BSs at time slot t . $\Lambda_{u,u'}^{\text{Sns}}(t) = 1$ means UAV u sense the state information of u' , and 0 if not.

3.1.3. Mobility model of MTs

We use a Gaussian-Markov random model [33] to simulate the movement properties of MTs, capturing their uncertain motion characteristics to better reflect real-world scenarios. At each time slot, a UAV-BS performs one service. The position of MT m at time slot t is denoted by $\mathbf{p}_m^{\text{MT}}(t) = [x_m^{\text{MT}}(t), y_m^{\text{MT}}(t), 0]^T$. The moving velocity and direction is denoted by $\mathbf{v}_m^{\text{MT}}(t)$ and $\theta_m^{\text{MT}}(t)$ as

$$\mathbf{v}_m^{\text{MT}}(t+1) = \eta_0 \mathbf{v}_m^{\text{MT}}(t) + (1 - \eta_0) \bar{\mathbf{v}} + \sqrt{1 - \eta_0^2} \gamma_v, \quad (2a)$$

$$\theta_m^{\text{MT}}(t+1) = \eta_1 \theta_m^{\text{MT}}(t) + (1 - \eta_1) \bar{\theta}_m + \sqrt{1 - \eta_1^2} \gamma_\theta, \quad (2b)$$

where $0 \leq \eta_0, \eta_1 \leq 1$ are used to adjust the effect of previous states. $\bar{\mathbf{v}}$ denotes the average velocity of all MTs, and $\bar{\theta}_m$ denotes the average

Table 2
Key notations in system model and problem formulation.

Notation	Description
W	Bandwidth of each subchannel (Hz)
$h_u^{UAV}(t)$	Flight altitude of UAV u (m)
$\mathbf{p}_u^{UAV}(t), \mathbf{p}_m^{MT}(t)$	Position of UAV u /MT m (m)
d_{min}	Safe distance between UAV-BSs (m)
$a_u^{UAV}(t)$	Acceleration of UAV u (m/s^2)
$v_u^{UAV}(t), v_m^{MT}(t)$	Velocity of UAV u /MT m (m/s)
$\theta_u^{UAV}(t), \theta_m^{MT}(t)$	Direction of UAV u /MT m ($^\circ$)
θ_{ts}	Antenna emission direction ($^\circ$)
$d_{u,m}(t)$	Distance between UAV u and MT m (m)
$g_{u,m}(t)$	Channel gain from UAV u to MT m
$R_u^{Sns}(t), R_u^{Cov}(t)$	Sensing/coverage radius of UAV u (m)
$\Omega^{Sns}(t), \Lambda^{Sns}(t)$	Sensing matrices of UAV-BSs
$\Phi^{Cov}(t)$	Coverage matrix of UAV-BSs
$\Gamma^{Sch}(t)$	Scheduling matrix of UAV-BSs
$T P_m^{MT}(t)$	Throughput of MT m (GB)
$E_u^{UAV}(t)$	Energy consumption of UAV u (KJ)

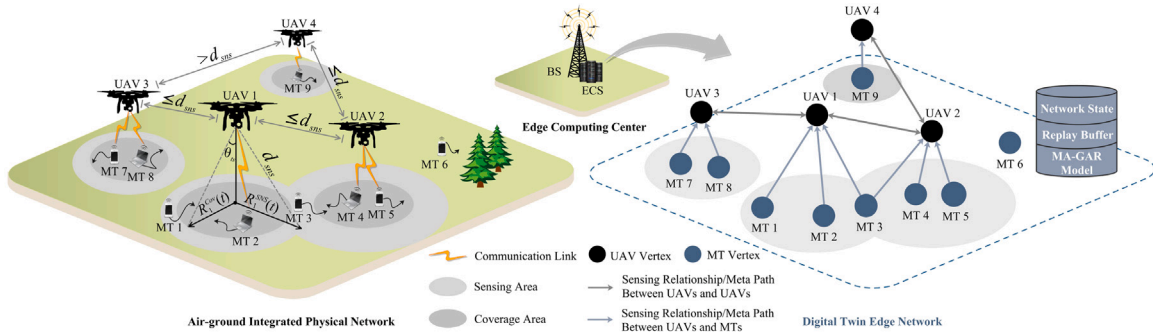


Fig. 1. Architecture of proposed air-ground integrated network. Notably, the initial sensing and coverage ranges of UAV-BSs are different and vary with flight altitude. The heterogeneous graph representation of air-ground integrated physical network topology is shown in digital twin edge network.

moving direction of MT m . Y_v and Y_θ are Gaussian perturbation terms of MTs in moving velocity and direction. The position of MT m can be updated by

$$\mathbf{p}_m^{MT}(t+1) = \begin{bmatrix} x_m^{MT}(t) + v_m^{MT}(t)\Delta t \cos(\theta_m^{MT}(t)) \\ y_m^{MT}(t) + v_m^{MT}(t)\Delta t \sin(\theta_m^{MT}(t)) \end{bmatrix}. \quad (3)$$

3.1.4. Air-ground channel model

The Air-to-Ground channel model has LoS or non-line-of-sight (NLoS) links. According to [38], the probability of LoS link from UAV u to MT m at time slot t is expressed as

$$\mathbf{P}_{LoS}(t) = \frac{1}{1 + \kappa_0 \exp[-\kappa_1(\theta_{u,m}(t) - \kappa_0)]}, \quad (4)$$

where κ_0 and κ_1 are environment-related constants. $\theta_{u,m}(t)$ denotes the elevation angle from u to m . $\theta_{u,m}(t) = \frac{180}{\pi} \arcsin(h_u^{UAV}(t)/d_{u,m}(t))$. $d_{u,m}(t)$ denotes the Euclidean distance from u to m . The probability of NLoS link from u to m is denoted by $\mathbf{P}_{NLoS}(t) = 1 - \mathbf{P}_{LoS}(t)$. The path losses for NoS/NLoS links are

$$\mathbf{L}_\zeta(t) = \left(\frac{4\pi f_c d_{u,m}(t)}{c} \right)^\lambda \eta_\zeta, \zeta \in \{LoS, NLoS\}, \quad (5)$$

where f_c denotes the center carrier frequency, c denotes the speed of light, λ denotes path loss exponent. η_{LoS} and η_{NLoS} denote the excessive path losses for LoS/NLoS links. The channel gain from UAV u to MT m at time slot t is

$$g_{u,m}(t) = \frac{1}{\mathbf{P}_{LoS}(t)\mathbf{L}_{LoS}(t) + \mathbf{P}_{NLoS}(t)\mathbf{L}_{NLoS}(t)}. \quad (6)$$

3.1.5. Communication model

The spectrum is divided into a set $\mathbb{C}$ of subchannels, each subchannel $c \in \mathbb{C} \triangleq \{1, 2, \dots, C\}$ with a fix bandwidth W . The communication

coverage matrix $\Phi^{Cov}(t) \in \{0, 1\}^{U \times M}$ describes the coverage situation of all UAV-BSs to all MTs at time slot t . $\Phi_{u,m}^{Cov}(t) = 1$ means MT m is within the coverage, and 0 means not. Thus, the signal power received by MT m from UAV u is denoted by $p_{ts} \cdot g_{u,m}(t)$. For interference between links, the signal-to-interference-plus-noise ratio of downlink from u to m is given by

$$\gamma_{u,m}(t) = \frac{\Phi_{u,m}^{Cov}(t)g_{u,m}(t)p_{ts}}{\sum_{u'=1, u' \neq u}^U \Phi_{u',m}^{Cov}(t)g_{u',m}(t)p_{ts} + W\sigma^2}, \quad (7)$$

where σ^2 denotes the power spectral density, p_{ts} is the transmission power.

When an MT is within the coverage of multiple UAV-BSs, it is preferentially scheduled to the UAV-BS with the nearest distance and available subchannels. Notably, if the number of MTs covered by a UAV-BS exceeds its available subchannels, the UAV-BS prioritizes MTs with lower average throughput. The scheduling matrix $\Gamma^{Sch}(t) \in \{0, 1\}^{U \times M \times C}$ represents the scheduling of all UAV-BSs for all MTs. $\Gamma_{u,m,c}^{Sch}(t) = 1$ means MT m is scheduled by UAV u with subchannel c . The throughput of MT m at time slot t can be estimated by

$$T P_m^{MT}(t) = \Delta t W \sum_{u=1}^U \sum_{c=1}^C \Gamma_{u,m,c}^{Sch}(t) \log(1 + \gamma_{u,m}(t)). \quad (8)$$

3.1.6. Energy consumption model of UAV-BSs

For UAVs, the energy consumption for communication is much lower than that for flight and can be ignored in this paper [39]. In this paper, we only consider the energy consumption required for flying. The initial battery energy of each UAV is fixed at $E_{batt}(J)$. According to [40], the propulsion energy consumption of UAV u at time slot t is estimated as

$$E_u^{UAV}(t) = \Delta t \left[\frac{\tau \rho s A \Delta t^3 a_u^{UAV}(t)^3}{128} + p_0 \left(1 + \frac{3 \Delta t^2 a_u^{UAV}(t)^2}{16 v_{tip}^2} \right) \right]$$

$$+ p_1 \left(\sqrt{1 + \frac{\Delta r^4 a_u^{\text{UAV}}(t)^4}{1024 v_{\text{rotor}}^4}} - \frac{\Delta r^2 a_u^{\text{UAV}}(t)^2}{32 v_{\text{rotor}}^2} \right)^{1/2} \Big], \quad (9)$$

where p_1 and p_0 denote the induced power and blade profile power in the hovering state, respectively. v_{tip} and v_{rotor} denote the tip velocity of the rotor blade and mean rotor induced velocity in hover, respectively. τ and ρ denote the fuselage drag ratio and air density, respectively. s and A denote the rotor solidity and disc area, respectively.

3.2. Problem formulation

Our objective is to achieve optimal downlink coverage for MTs while guaranteeing the energy efficiency of UAV-BSs. We divide it into two subproblems: $\mathcal{P}_1$ and $\mathcal{P}_2$. The objective of $\mathcal{P}_1$ is to jointly optimize the scheduling matrix $\Gamma^{\text{Sch}}(t)$, positions $\mathbb{P} \triangleq \{\mathbf{p}_u^{\text{UAV}}(t), \forall u \in \mathbb{U}, \forall t \in \mathbb{T}\}$ and velocities $\mathbb{V} \triangleq \{v_u^{\text{UAV}}(t), \forall u \in \mathbb{U}, \forall t \in \mathbb{T}\}$ of UAV-BSs to maximize MTs' throughput. The objective of $\mathcal{P}_2$ is to minimize the energy consumption of all UAVs. The mathematical descriptions of these two subproblems are described as follows.

Constraints (10a) and (10b) ensure that UAV-BSs and MTs remain within the ground area. Constraint (10c) limits the altitude range of UAV-BSs. (10d) limits the maximum velocity of UAV-BSs. Constraint (10e) enforces a minimum safe distance between UAV-BSs to prevent collisions. (10f) represents the battery limitation of UAV-BSs. (10g) ensures that each MT occupies at most one subchannel of a UAV-BS simultaneously. (10h) ensures that each MT is scheduled by at most one UAV-BS simultaneously.

It can be readily derived that problems $\mathcal{P}_1$ and $\mathcal{P}_2$ are mixed-integer nonlinear programming (MINLP) problems with a large number of variable couplings. In the next section, we will address these two problems simultaneously.

$$\mathcal{P}_1 : \max_{\{\Gamma^{\text{Sch}}(t), \mathbb{P}, \mathbb{V}\}} \sum_{t=1}^T \sum_{m=1}^M T P_m^{\text{MT}}(t)$$

$$\mathcal{P}_2 : \min_{\{\Gamma^{\text{Sch}}(t), \mathbb{P}, \mathbb{V}\}} \sum_{t=1}^T \sum_{u=1}^U E_u^{\text{UAV}}(t)$$

$$\text{s.t. } 0 \leq x_u^{\text{UAV}}(t), x_m^{\text{MT}}(t) \leq L, \forall u, \forall m, \forall t, \quad (10a)$$

$$0 \leq y_u^{\text{UAV}}(t), y_m^{\text{MT}}(t) \leq L, \forall u, \forall m, \forall t, \quad (10b)$$

$$h_{\min}^{\text{UAV}} \leq h_u^{\text{UAV}}(t) \leq h_{\max}^{\text{UAV}}, \forall u, \forall t, \quad (10c)$$

$$a_u^{\text{UAV}}(t) \leq a_{\max}^{\text{UAV}}, \forall u, \forall t, \quad (10d)$$

$$\|p_i^{\text{UAV}}(t) - p_j^{\text{UAV}}(t)\|_2 \geq d_{\min}, \forall i, \forall j, i \neq j, \forall t, \quad (10e)$$

$$\sum_{t=1}^T E_u^{\text{UAV}}(t) \leq E_{\text{batt}}, \forall u, \quad (10f)$$

$$\sum_{m=1}^M \Gamma_{u,m,c}^{\text{Sch}}(t) \leq 1, \forall u, \forall c, \forall t, \quad (10g)$$

$$\sum_{u=1}^U \sum_{c=1}^C \Gamma_{u,m,c}^{\text{Sch}}(t) \leq 1, \forall m, \forall t. \quad (10h)$$

4. The energy-efficiency trajectory planning framework based multi-agent graph reinforcement learning

In this section, we first give the POMDP formulation, then we introduce the details of the proposed MA-GAR framework.

4.1. POMDP formulation

Each UAV-BS has an agent, and the set of agents is denoted by $\mathbb{U}$. The environment is the whole air-ground integrated network. Its state space is denoted by $\mathcal{S} \triangleq \{s(t), \forall t \in \mathbb{T}\}$. The local observation space of each agent is denoted by $\mathcal{O}_u \triangleq \{o_u(t), \forall t \in \mathbb{T}\}$. The action space is denoted by $\mathcal{A}_u \triangleq \{a_u(t), \forall t \in \mathbb{T}\}$. Each agent interacts with

the environment at discrete time steps. At time step t , UAV u obtains a local observation $o_u(t)$ of the current environment state $s(t)$, where $o_u(t) \subseteq s(t)$. Then executes action $a_u(t)$ according to the learned strategy and obtains a fresh observation $o_u(t+1)$ and a reward $r_u(t)$. The details are given below.

4.1.1. Observation space

The local observation o_u^{Self} of UAV u includes the UAV's position, the number of occupied subchannels, and the remaining energy, expressed as follows:

$$o_u^{\text{Self}}(t) = \left\{ \mathbf{p}_u^{\text{UAV}}(t), \sum_{m=1}^M \sum_{c=1}^C \Gamma_{u,m,c}^{\text{Sch}}(t), E_u^{\text{Batt}}(t) \right\}, \quad (11)$$

$$o_u^{\text{UAVs}}(t) = \left\{ \mathbf{p}_{u'}^{\text{UAV}}(t), \sum_{m=1}^M \sum_{c=1}^C \Gamma_{u',m,c}^{\text{Sch}}(t), E_{u'}^{\text{Batt}}(t) \right\}_{u' \in \mathbb{U}_u^{\text{Sns}}(t)}, \quad (12)$$

$$o_u^{\text{MTs}}(t) = \left\{ \mathbf{p}_m^{\text{MT}}(t), T P_m^{\text{MT}}(t), \sum_{c=1}^C \Gamma_{u,m,c}^{\text{Sch}}(t), \sum_{u=1}^U \sum_{c=1}^C \Gamma_{u,m,c}^{\text{Sch}}(t), \Phi_{u,m}^{\text{Cov}}(t), I \left(\sum_{u=1}^U \Phi_{u,m}^{\text{Cov}}(t) > 0 \right) \right\}_{m \in \mathbb{M}_u^{\text{Sns}}(t)}. \quad (13)$$

Besides the local observation, UAV u can also get the information of other UAVs' observation $o_{u'}^{\text{UAVs}}$ and MTs' observation o_m^{MTs} within its sensing area. We use the sets $\mathbb{U}_u^{\text{Sns}}(t), \mathbb{M}_u^{\text{Sns}}(t)$ to denote all other UAVs and MTs sensed by the UAV u at time step t . The type of observations for o_u^{UAVs} is the same as that for $o_u^{\text{Self}}(t)$, as shown in Eq. (12). o_u^{MTs} includes MT m 's position, throughput, whether it is scheduled by the current agent, whether it is scheduled by other UAV agents, whether it is within the coverage area of the current UAV, and whether it is within the coverage area of other UAV agents. It is given by Eq. (13), where $I(\cdot)$ is an indicator function. The total observation of UAV u at time step t is summarized as $o_u(t) = o_u^{\text{Self}}(t) \cup o_u^{\text{UAVs}}(t) \cup o_u^{\text{MTs}}(t)$, which is obtained through collaboration among UAVs and MTs. Additionally, the sensing area of UAV agents varies with flight altitude, resulting in observations of different dimensions at each time step.

4.1.2. Action space

Since the optimization objective is particularly sensitive to the flight trajectory, it is necessary to design continuous control actions to finely adjust the UAVs' trajectories. At time step t , the flight action of UAV u includes its flight direction and acceleration in 3D space, and is denoted by

$$a_u(t) = \{\theta_u^{\text{UAV,H}}(t), \theta_u^{\text{UAV,V}}(t), a_u^{\text{UAV}}(t)\}. \quad (14)$$

4.1.3. Reward function

In order to further solve the restriction of local observation information, rewards are divided into a global term and an individual term. First, UAV-BSs synchronize their interaction experiences and state through DITEN of the ECC at every time slot. Then the ECC calculates the global reward term $r_G(t)$ to further promote the cooperation of agents as

$$r_G(t) = \alpha \beta_0 \sum_{m=1}^M T P_m^{\text{MT}}(t) - (1 - \alpha) \beta_1 \sum_{u=1}^U E_u^{\text{UAV}}(t), \quad (15)$$

where β_0 and β_1 are factors that normalize throughput and energy consumption to the same scale. $0 \leq \alpha \leq 1$ denotes the weight coefficient for throughput and energy consumption, which can be tested and adjusted in practice based on the actual conditions of the scenarios. After receiving the round of global reward terms, each UAV agent calculates its own final reward $r_u(t)$. To prevent catastrophic events, we introduce a penalty mechanism into the final reward function. At time step t , it is defined as

$$r_u(t) = \gamma r_G(t) - \varphi_u(t) \mathcal{P}, \quad (16)$$

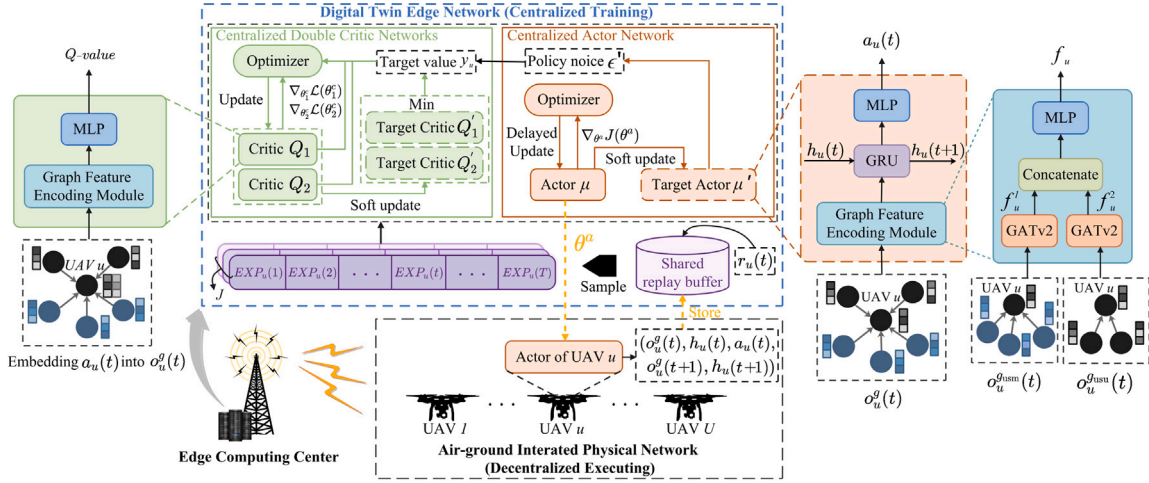


Fig. 2. Architecture of proposed multi-agent graph attention recurrent actor-critic framework with DT-empowered CTDE mechanism. The left is multi-agent centralized double critic networks. The right is recurrent multi-agent actor network and graph feature encoding module.

where γ is the reward scaling factor, $\phi_u(t)$ denotes the number of mistakes made by UAV u at time step t . There are three types of mistakes: crossing the area boundary, violating the safety distance, and low power. Whenever an agent violates any penalty mechanisms, it will be punished with a fixed-size penalty $\mathcal{P}$.

4.2. Multi-agent graph attention recurrent actor-critic framework with DT-empowered CTDE mechanism

In this subsection, we first introduce a graph representation method for the relationships between network entities. Then we introduce the key components of MA-GAR, including a recurrent multi-agent actor network, centralized double critic networks, and a graph feature encoding module with a multi-head attention mechanism. Finally, we describe in detail the DT-empowered CTDE mechanism of MA-GAR. Fig. 2 shows the overall architecture of MA-GAR with the DT-empowered CTDE mechanism.

4.2.1. Heterogeneous graph representation for the relationships between UAV-BSs and network entities

We use a heterogeneous graph $H \triangleq \{\mathcal{V}, \mathcal{E}\}$ to represent the environment information of the network, where $\mathcal{V}$ is the set of vertices, $\mathcal{E}$ denotes the set of edges. $\mathcal{V}$ contains two types: UAVs and MTs, i.e. $\mathcal{V} = \mathcal{V}_{\text{UAV}} \cup \mathcal{V}_{\text{MT}}$. As illustrated in the right part of Fig. 1, we define two meta-paths in the heterogeneous graph based on the sensing relationships between UAV-BSs and other network entities: (1) MT to UAV; (2) UAV to UAV. Each edge is denoted by $e = (v_1, v_2)$, where v_1 is the source vertex and v_2 is the destination vertex, and $\forall v \in \mathcal{V}, \forall e \in \mathcal{E}$. Please note that since UAVs are agents, the type of all the destination vertices is UAVs. Edge (u', u) means that UAV u' is within the sensing area of UAV u . Similarly, Edge (m, u) means that MT m is within the sensing area of UAV u .

Since each UAV agent only obtains partially observable information about the environment, its observation is a local subgraph of the heterogeneous graph H . The agent's observation $o_u(t)$ at time step t is converted into a heterogeneous graph representation $o_u^g(t)$, which can be further decomposed into a unidirectional bipartite graph containing two meta-paths, $(MT, \text{sensed_by_UAV})$ and $(UAV, \text{sensed_by_UAV})$, denoted by $o_u^{\text{msu}}(t)$ and $o_u^{\text{usu}}(t)$, which is the input of the graph feature encoding module of the framework.

4.2.2. Recurrent multi-agent actor network on UAVs

Each UAV agent is equipped with an actor network consisting of a gated recurrent unit (GRU) and the graph feature encoding module, as shown on the right of Fig. 2. The details of the graph feature encoding module will be discussed later. The DITEN on ECC contains a replay buffer $\mathcal{B}$ shared by all UAV agents. To ensure order and correlation among samples participating in network update, we design an experience storage mechanism for the replay buffer. The experience generated by UAV u at time step t is $EXP_u(t) = (o_u^g(t), h_u(t), a_u(t), r_u(t), o_u^g(t+1), h_u(t+1))$, where $h_u(t)$ is the hidden layer vector of the actor network of UAV u . We stack the complete interaction experience of each UAV agent within an episode into a transition in the replay buffer, denoted by $\{EXP_u(t)\}_{t \in \mathcal{T}, \forall u \in \mathcal{U}}$. DITEN synchronizes physical network states and integrates real-time experiences during training, storing them in the replay buffer according to the described storage mechanism.

Besides the actor network $\mu_u(o_u^g(t), h_u(t); \theta^a)$ for UAV u , we also define a centralized actor network $\mu(o_u^g(t), h_u(t); \theta^a)$ and a centralized actor-target network $\mu'(o_u^g(t+1), h_u(t+1); \theta^{a'})$, where θ^a and $\theta^{a'}$ denote the parameters of the actor network and the target network, respectively. Since we adopt a centralized training mechanism, all UAV actor networks share a single network parameter θ^a . To facilitate exploration, we add noise to the action output. Thus, the action is defined as

$$a_u(t) = \text{clip}[\mu_u(o_u^g(t), h_u(t); \theta^a) + \epsilon, -1, 1], \quad (17)$$

where ϵ denotes exploratory noise generated from a Gaussian distribution, $\epsilon \sim \mathcal{N}(0, \sigma_\mu)$, and σ_μ controls the degree of discretization of the exploration noise. The clip range is $[-1, 1]$ to keep the action value within a normal range after adding noise.

In addition, to improve training stability, we use a delayed policy to update the actor network at a lower frequency D_{update} . The training of the centralized actor network is on the DITEN, where a random mini-batch of J transitions is sampled from the replay buffer $\mathcal{B}$ for training. The centralized actor network is updated using deterministic policy gradients of Q -values, and the update process is given by

$$\begin{aligned} \nabla_{\theta^a} J(\theta^a) &= \frac{1}{J \cdot T} \sum_{j=1}^J \sum_{t=1}^T \nabla_{\theta^a} \mu(o_u^g(t), h_u(t); \theta^a) \cdot \nabla_{a_u(t)} Q_1(o_u^g(t), a_u(t); \theta^c) \Big|_{a_u(t)}, \end{aligned} \quad (18)$$

where $a_u(t) = (o_u^g(t), h_u(t); \theta^a)$.

Finally, we update the centralized actor network's parameters according to Eq. (18) and update the actor target network with a soft update strategy, as shown below.

$$\theta^a \leftarrow \theta^a - \partial_a \nabla_{\theta^a} J(\theta^a), \quad (19)$$

$$\theta^{a'} = \varsigma \theta^a + (1 - \varsigma) \theta^{a'}, \quad (20)$$

where θ_a is the learning rate and ς is the update rate of the target network.

4.2.3. Multi-agent centralized double critic networks

To avoid overestimating Q -value with a single critic network, we adopt a structure with centralized double critic networks to prevent over-valuation deviation in the DITEN. As shown in Fig. 2 the centralized double critic networks are $Q_1(o_u^g(t), a_u(t); \theta_1^c)$ and $Q_2(o_u^g(t), a_u(t); \theta_2^c)$. The two target networks are $Q_1'(o_u^g(t+1), a_u'(t+1); \theta_1^{c'})$ and $Q_2'(o_u^g(t+1), a_u'(t+1); \theta_2^{c'})$. θ_1^c, θ_2^c and $\theta_1^{c'}, \theta_2^{c'}$ correspond to the parameters of the double critic networks and target networks, respectively. $a_u'(t+1)$ is the target action after policy smoothing. Specifically, we perturb the target action by adding random noise to smooth the target action, which is given by

$$a_u'(t+1) = \mu'(o_u^g(t+1), h_u(t+1); \theta^{a'}) + \epsilon', \quad (21)$$

where ϵ' is policy noise generated from a Gaussian distribution with $\epsilon' \sim \text{clip}(\mathcal{N}(0, \sigma_{\mu'}), -\rho, \rho)$. The constant ρ ensures stability by clipping noise.

The centralized double critic networks are trained in the DITEN. Similar to the updating process of the actor network, we randomly sample J mini-batch transitions from B . The critic network adopts a policy gradient approach with a mean squared error loss function to minimize the difference between the target and predicted Q -value. The loss is defined as

$$\mathcal{L}(\theta_i^c) = \frac{1}{J \cdot T} \sum_{j=1}^J \sum_{t=1}^T \left[y_u(t) - Q_i(o_u^g(t), a_u(t); \theta_i^c) \right]^2, \quad (22)$$

where $i \in \{1, 2\}$. $y_u(t)$ is the target Q -value which is calculated as

$$y_u(t) = r_u(t) + \epsilon_r \cdot \min \left[Q_1'(o_u^g(t+1), a_u'(t+1); \theta_1^{c'}), Q_2'(o_u^g(t+1), a_u'(t+1); \theta_2^{c'}) \right], \quad (23)$$

where ϵ_r is the reward discount factor. Then we use the minimum value of Q_1', Q_2' as the target Q -value.

Finally, the double critic networks' parameters are updated according to Eq. (22) and the target networks are updated with a soft update strategy, as shown below.

$$\theta_i^c \leftarrow \theta_i^c - \partial_c \nabla_{\theta_i^c} \mathcal{L}(\theta_i^c), \quad i \in \{1, 2\}, \quad (24)$$

$$\theta_i^{c'} = \varsigma \theta_i^c + (1 - \varsigma) \theta_i^{c'}, \quad i \in \{1, 2\}, \quad (25)$$

where ∂_c is learning rate of the critic networks.

4.2.4. Graph feature encoding module with multi-head attention mechanism for actor-critic network

In the graph feature encoding module, the multi-head attention mechanism [41] is introduced to adaptively capture the underlying relationships between UAV-BSs and network entities (MTs or other UAV-BSs), helping it to allocate more attention to more valuable entities, thereby enhancing its flight strategy. As shown on the right of Fig. 2, for each subgraph $o_u^{\text{gusm}}(t)$ and $o_u^{\text{gusu}}(t)$, we use the self-attention mechanism on UAV vertices to assign weights to source vertices, i.e., sensed MTs or other UAVs. The feature vector of MT m at time step t is defined as $\mathbf{f}_m^{\text{MT}}(t)$, where $\forall m \in \mathbb{M}_u^{\text{Sns}}(t)$. The feature vector of UAV u' is defined as $\mathbf{f}_{u'}^{\text{UAV}}(t)$, where $\forall u' \in \mathbb{U}_u^{\text{Sns}}(t)$. The current features of UAV u in the two subgraphs are defined as $\mathbf{f}_u^1$ and $\mathbf{f}_u^2$, with initial value $\mathbf{f}_u^{\text{UAV}}(t)$. Thus, the attention values between UAV vertices and source vertices in subgraphs $o_u^{\text{gusm}}(t)$ and $o_u^{\text{gusu}}(t)$ are calculated as

$$e_{u,m} = \mathbf{a}_1 \cdot \text{LeakyReLU}(\mathbf{W}_1 \cdot [\mathbf{f}_u^1 \parallel \mathbf{f}_m^{\text{MT}}(t)]), \quad (26)$$

$$e_{u,u'} = \mathbf{a}_2 \cdot \text{LeakyReLU}(\mathbf{W}_2 \cdot [\mathbf{f}_u^2 \parallel \mathbf{f}_{u'}^{\text{UAV}}(t)]), \quad (27)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are trainable vectors, which are mainly used for linear transformation of features. $\parallel$ denotes the concatenation operation.

$\mathbf{a}_1, \mathbf{a}_2$ are multilayer perceptrons (MLPs) used to calculate the similarity of vectors. Then, we use the softmax function to normalize the attention coefficients to make them easily comparable among different vertices as

$$\alpha_{u,m} = \text{softmax}(e_{u,m}) = \frac{\exp(e_{u,m})}{\sum_m \mathbb{M}_u^{\text{Sns}}(t) \exp(e_{u,m})}, \quad (28)$$

$$\alpha_{u,u'} = \text{softmax}(e_{u,u'}) = \frac{\exp(e_{u,u'})}{\sum_{u'} \mathbb{U}_u^{\text{Sns}}(t) \exp(e_{u,u'})}. \quad (29)$$

To improve the expressiveness and stability of the model, we adopt the multi-head attention mechanism, where K independent attention mechanisms are performed for each UAV vertex. The output feature representations of subgraphs $o_u^{\text{gusm}}(t)$ and $o_u^{\text{gusu}}(t)$ are given by

$$\mathbf{f}_u^1 = \left\|_k \sigma_1 \left(\sum_m \mathbb{M}_u^{\text{Sns}}(t) \alpha_{u,m} \mathbf{W}_1^{\text{Agent}} \mathbf{f}_m^{\text{MT}}(t) \right), \quad (30)$$

$$\mathbf{f}_u^2 = \left\|_k \sigma_2 \left(\sum_{u'} \mathbb{U}_u^{\text{Sns}}(t) \alpha_{u,u'} \mathbf{W}_2^{\text{Agent}} \mathbf{f}_{u'}^{\text{UAV}}(t) \right), \quad (31)$$

where $\mathbf{W}_1^{\text{Agent}}, \mathbf{W}_2^{\text{Agent}}$ are the corresponding learnable weight matrices, σ_1, σ_2 are nonlinear functions (typically ReLU), and k denotes the execution of the k th attention mechanism. Finally, a linear transformation is performed for $\mathbf{f}_u^1 \parallel \mathbf{f}_u^2$ by using an MLP, then the final encoding output of the graph feature encoding module is generated by $\mathbf{f}_u$.

We also design a graph feature encoding module for the centralized double critic networks, as shown on the left of Fig. 2. This module is similar to the actor network, except that the graph feature encoding module of the double critic networks must first embed the agent's action $a_u(t)$ into the UAV vertex in graph $o_u^g(t)$.

4.2.5. DT-empowered CTDE mechanism of MA-GAR

A decentralized independent training mechanism usually results in non-smooth and inefficient learning processes. Additionally, due to the energy restriction, training on UAVs is not a better choice. To cope with these problems, we introduce a DT-empowered CTDE mechanism in MA-GAR, as shown in Fig. 2. During the training phase, interaction experiences of UAVs of the physical network are synchronized to the DITEN, which then further integrates experiences and stores them into the replay buffer. When sufficient transitions are accumulated, the DITEN updates the parameters of the centralized actor network and the centralized double critic networks. In execution phase, the updated centralized actor network is downloaded to all UAVs. Algorithm 1 gives the details of the process.

5. Numerical results and analysis

In this section, we comprehensively verify the performance of our proposed MA-GAR through extensive numerical simulations. The compared benchmarks are listed as follows:

- 1 GVis&Comm [26]: It is a GNNs-based MARL method. It uses a Q-Learning-based algorithm and facilitates discrete information communication among agents by leveraging GNNs to process a heterogeneous graph, without restricting communication distance.
- 2 MADDPG [34]: It is an off-policy algorithm using deterministic policy gradients and is one of the mainstream MARL algorithms.
- 3 MADQN [42]: It estimates Q-values using deep neural networks based on Q-Learning to achieve learning of agents.
- 4 Randomized: UAV agents execute random actions during each interaction.

Algorithm 1: MA-GAR with DT-Empowered CTDE

```

1 Input: Graph information observed by UAVs, model and
   environment parameters, training hyperparameters
2 Output: Energy efficiency trajectory planning strategy for
   UAV-BSs
3 Initialize the centralized actor network  $\mu$ , the centralized
   double critic networks  $Q_1, Q_2$ , the target networks and the
   replay buffer  $\mathcal{B}$  in the DITEN;
4 Initialize UAV agents' actor networks  $\mu_u, \forall u \in \mathcal{U}$  in the
   air-ground integrated physical network;
5 for  $epoch \leftarrow 1$  to  $N_{epoch}$  do
6   for  $episode \leftarrow 1$  to  $N_{episode}$  do
7     for  $t \leftarrow 1$  to  $T$  do
8       for each UAV agent  $u$  do
9         Obtain observation  $o_u(t)$ , represent it as graph
            $o_u^g(t)$ , and acquire hidden layer vector  $h_u(t)$ ;
10        Execute action  $a_u(t)$ , obtain  $o_u(t+1)$ , represent it
           as  $o_u^g(t+1)$ , and acquire  $h_u(t+1)$ ;
11        Integrate the partial interaction experience and
           synchronize it to DITEN;
12        DITEN evaluates reward  $r_u(t)$ , integrates it into
           the complete experience  $EXP_u(t)$ , and store it
           into  $\mathcal{B}$ ;
13      end
14    end
15    if  $|\mathcal{B}| \geq J$  then
16      Sample  $J$  mini-batch transitions from  $\mathcal{B}$ ;
17      Update the centralized double critic networks
        according to (22) and (24);
18      if  $episode \bmod D_{update} = 0$  then
19        Update the centralized actor network according
          to (18) and (19);
20        Update the target networks with a soft update
          strategy as described in (20) and (25);
21      end
22    end
23    The updated centralized actor network is downloaded
      to all UAV agents in the physical network;
24  end
25 end

```

5.1. Simulation settings

Since the existing algorithms either do not consider performance or are general RL-based algorithms. To more reasonably compare the performance of all algorithms, we replace the reward function of the compared four benchmark algorithms with the one used in our proposed MA-GAR defined by Eq. (16), which includes the impact of energy consumption. The simulation experiments are based on Python 3.10, PyTorch 2.1.2, DGL 2.3.0, and the server with Ubuntu 22.04. The detailed configuration of the server includes an a 48-core AMD EPYC 7642 CPU, two NVIDIA GeForce RTX 3090 GPUs with 24 GB VRAM, and 80 GB RAM. The initial positions of MTs and UAV-BSs are randomly distributed in a ground area of 1600 m $\times$ 1600 m. UAV-BSs are constrained to an altitude range of 50 m to 150 m, with an initial flight altitude of 100 m. MT's movement defaults to the Gaussian-Markov random model. Table 3 summarizes additional key environment parameters.

To provide more concise results, we train the proposed MA-GAR for $N_{epoch} = 200$ epochs, each epoch containing $N_{episode} = 20$ episodes, and each episode consisting $T = 200$ time steps. Mini-batch size is $J = 32$. Replay buffer capacity is $|\mathcal{B}| = 5000$. Reward discount factor is $\epsilon_r = 0.99$.

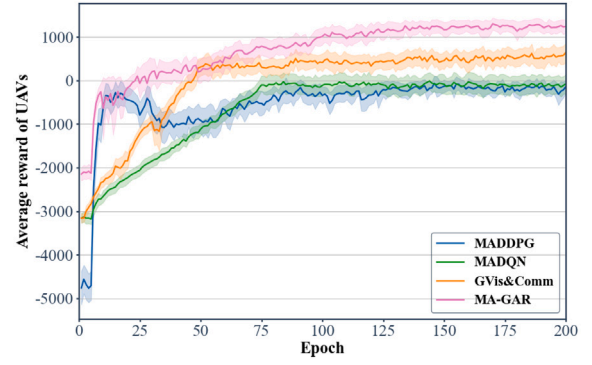


Fig. 3. Convergence of UAV agents' reward ($U = 4, M = 10$).

Actor network's learning rate is $\vartheta_a = 5 \times 10^{-5}$. Critic networks' learning rate is $\vartheta_c = 5 \times 10^{-4}$. Update rate of the target network is $\zeta = 5 \times 10^{-3}$. Actor network update frequency is $D_{update} = 2$. The number of attention heads is $K = 4$. Additionally, to address data randomness, we set the random seeds to 11–20 across all experiments. Each experiment was run 10 times to evaluate robustness and ensure statistically reliable conclusions. In simulation figures, the solid lines represent the mean values, while the shadows/error bars indicate the standard error.

5.2. Convergence analysis

We first evaluate the convergence of UAV agents' reward in a simulation with 4 UAVs and 10 MTs ($U = 4, M = 10$). As we discussed in our simulation settings, we calculate the average reward of all UAVs over all episodes within each epoch, and the results are shown in Fig. 3. From the results, we can find that the average reward of our proposed MA-GAR and all benchmarks gradually increases with the number of training epochs, then eventually reaches a relatively stable value. The final converged reward value of MA-GAR is considerably higher than GVis&Comm, MADQN, and MADDPG. Furthermore, MA-GAR's average reward is higher than benchmarks at the beginning and goes up faster, indicating that MA-GAR has higher learning efficiency in the early stages of training. MA-GAR reaches a level comparable to the final converged reward values of MADQN and MADDPG at about 6 epochs and reaches a level comparable to GVis&Comm at about 30. Subsequently, the reward value of MA-GAR increases steadily until it reaches the final convergence value at about 100 epochs, which is higher than GVis&Comm. This is because our proposed MA-GAR enables UAV agents to learn more optimal trajectory planning strategies than compared algorithms. Through the experiment, MADDPG exhibits fluctuations and large shadows due to the extreme randomness and dynamics, partially observable features of the environment. In contrast, MA-GAR has a smoother curve with smaller shadows. Thus, it enables UAVs to make accurate and efficient flight decisions in such complex environments.

Subsequently, Fig. 4 shows the convergence of average total throughput. Note that the system total throughput here refers to the sum of all UAV-BSs' and MTs' throughput, where the latter is calculated according to Eq. (8); the former is the accumulation of throughput of MTs scheduled on their occupied subchannels. The experimental results show that the throughput for all algorithms shows a steady upward trend at the beginning. Although MADDPG converges faster initially, it experiences a decline in performance, with the most severe fluctuations among all algorithms in long-term running. Ultimately, MADQN, MADDPG, and GVis&Comm achieve about 75 GB, 90 GB, and 160 GB of system total throughput, respectively, while our proposed MA-GAR achieves approximately 240 GB throughput. In terms of the upper bound on the maximum throughput that can be achieved, our proposed MA-GAR achieves an increase of about 191% compared to

Table 3
Key environment parameters.

Parameters	Values (Unit)
Duration of service period	2000 (s)
Duration of each time slot Δt	10 (s)
Maximum acceleration of flight a_{max}^{UAV}	4 (m/s ²)
Minimum safe distance d_{min}	10 (m)
Antennas angle θ_{is}	60 (°)
Sensing distance d_{sns}	412 (m)
Channel settings κ_0, κ_1	9.61, 0.16
Center carrier frequency f_c	2.4e9 (Hz)
Path loss exponent λ	2
Excessive path losses η_{LoS}, η_{NLoS}	1, 20 (dB)
Number of subchannels C	10
Bandwidth of each subchannel W	1 (MHz)
Power spectral density σ^2	-170 (dBm/Hz)
Transmission power p_{ts}	0.5 (Watt)
Initial battery energy E_{batt}	918074 (KJ)
UAV settings P_0, P_1	79.86, 88.63 (Watt)
UAV settings v_{tip}, v_{rotor}	120, 4.03 (m/s)
UAV settings τ, s, A	0.6, 0.05, 0.503

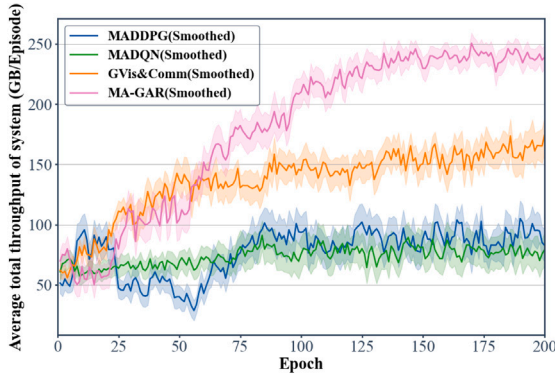


Fig. 4. Convergence of system total throughput ($U = 4, M = 10$).

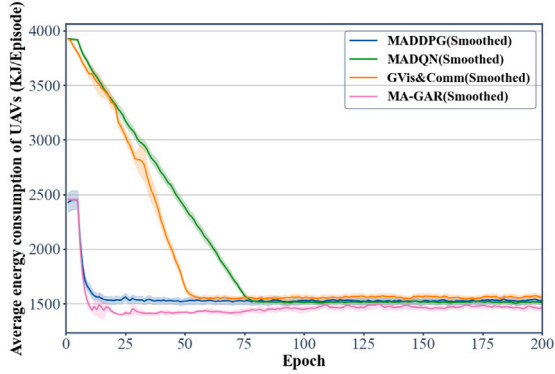


Fig. 5. Convergence of energy consumption ($U = 4, M = 10$).

MADQN and MADDPG and about 50% compared to GVis&Comm. The UAV-BSS trained by MA-GAR provide more stable service and deliver the best service quality to MTs.

The next experiment verifies the convergence of the average energy consumption of UAVs. As shown in Fig. 5, the results indicate that our proposed MA-GAR converges fastest among the four algorithms. MA-GAR and MADDPG converge at about 10 epochs and 20 epochs, while GVis&Comm and MADQN converge at about 60 epochs and 80 epochs, respectively. Ultimately, MADQN, MADDPG, and GVis&Comm achieve about 1510 KJ, 1530 KJ, and 1560 kJ of energy consumption, respectively, while our proposed MA-GAR achieves approximately 1465 kJ of energy consumption. Because MA-GAR enables UAV-BSS to learn

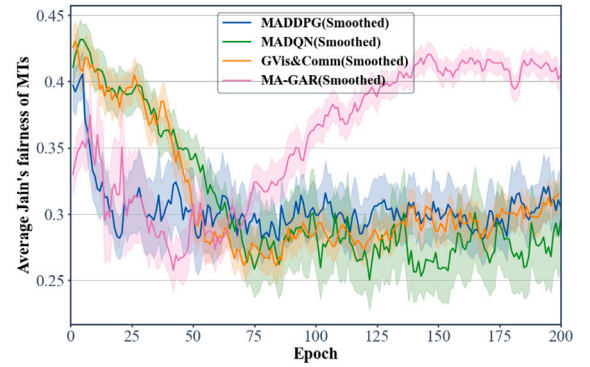


Fig. 6. Convergence of service fairness ($U = 4, M = 10$).

more energy-efficient strategies quickly, it verifies its effectiveness in energy-efficient trajectory planning. In addition, combining the results in Figs. 3 to 5, we can see that our proposed MA-GAR algorithm outperforms the other compared algorithms in reward, throughput, and energy consumption.

To verify the fairness of algorithms, we use the fairness criterion in [43] as $J(t) = \frac{(\sum_{m=1}^M TP_m^{MT}(t))^2}{M \sum_{m=1}^M TP_m^{MT}(t)^2}$. Fig. 6 shows the convergence of the fairness of MTs. From the results, we found that the fairness of MADDPG, MA-GAR, GVis&Comm, and MADQN shows a downward trend in the first 12, 40, 60, and 75 epochs, respectively. The all benchmark algorithms tend to be stable after the decline and fluctuate within a range, and GVis&Comm has a slight upward trend. However, our proposed MA-GAR can increase significantly and stabilize in the higher fairness region in subsequent epochs. The reason for the initial decline in fairness is that UAVs can initially cover and serve MTs around it. However, when the trajectory planning strategy is not good enough, the fairness of their initial serviceability will be reduced. In other words, the UAV only senses the nearby MT and lacks good mobility decision-making ability. However, with the convergence of the algorithm, the proposed MA-GAR shows an obvious increase, indicating that agents have explored a better trajectory planning strategy, which makes UAVs try their best to cover more MTs. The convergence experiments done so far demonstrate that our proposed MA-GAR can not only enable UAV-BSS to achieve the highest throughput at minimal energy cost but also provide a more fair service for MTs.

Fig. 7 shows the convergence of the average number of penalties for UAV agents. As shown in the results, the penalty number for UAV agents trained by GVis&Comm, MADDPG, and MADQN converges at

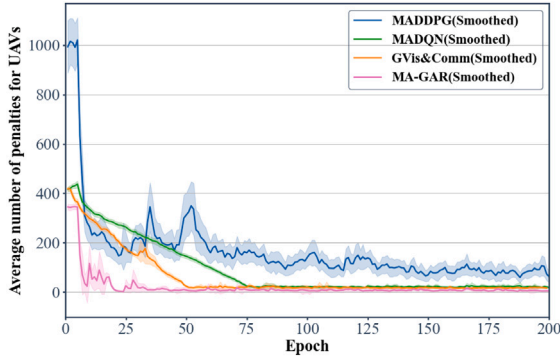


Fig. 7. Convergence of penalty number for UAV agents ($U = 4, M = 10$).

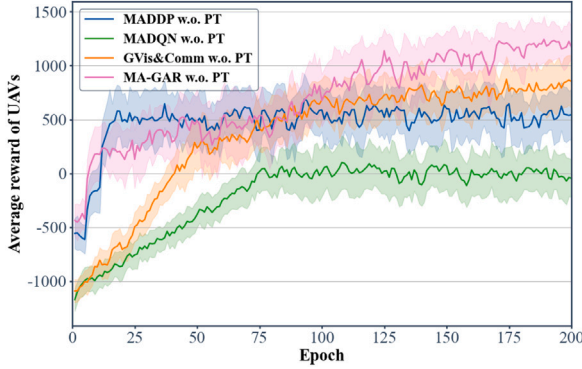


Fig. 8. Convergence of UAV agents' reward without penalty terms ($U = 4, M = 10$).

about 50, 60, and 80 epochs, respectively, while our proposed MA-GAR converges at about 25 epochs. Our proposed MA-GAR remains at the lowest level throughout the training process. These results indicate that our designed penalty mechanism performs best with MA-GAR. It can minimize catastrophic errors and provide higher safety and reliability for UAV-BSSs.

Fig. 8 illustrates the convergence of the UAV agent's rewards without the penalty term (PT). As shown in the results, MA-GAR rapidly converges to a high value during the early training phase (at 6 epochs) and maintains a stable upward trend thereafter. Ultimately, the average reward value at which MA-GAR converges is higher than that of all benchmark algorithms. This result demonstrates that, even in a "reward-only" environment without PT, MA-GAR can effectively drive UAV agents to develop robust and energy-efficient trajectory planning strategies.

Fig. 9 illustrates the convergence of the average reward of MA-GAR's UAV agents under different reward discount factor ϵ_r settings (0.85, 0.9, 0.95, and 0.99). It can be observed from the figure that the performance of MA-GAR gradually improves as ϵ_r increases. Specifically, when ϵ_r is 0.85 or 0.9, the algorithm converges faster but achieves lower final reward values. In contrast, when ϵ_r is 0.95 or 0.99, although convergence is slower, higher cumulative rewards are attained. This indicates that a lower ϵ_r makes MA-GAR more focused on immediate rewards, thereby accelerating convergence. On the other hand, a higher ϵ_r encourages MA-GAR to prioritize long-term cumulative rewards, resulting in better overall performance. These results validate the importance of long-term planning in optimizing energy-efficient trajectory strategies for UAVs within our MA-GAR framework.

Fig. 10 presents the reward convergence results of MA-GAR under four MT mobility models: the directed mobility model with a random initial direction of each MT (DMM-E), the directed mobility model with a random initial direction of all MTs (DMM-A), the random roaming

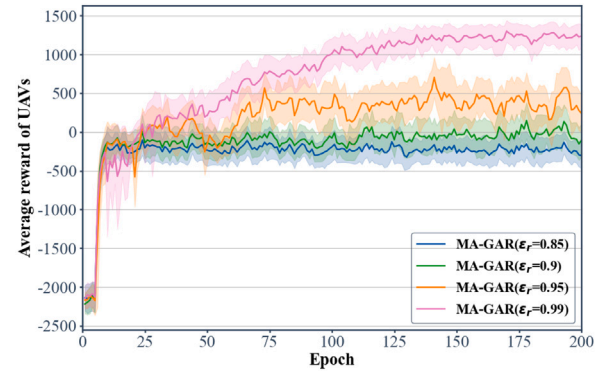


Fig. 9. Convergence of UAV agents' reward with different reward discount factor ($U = 4, M = 10$).

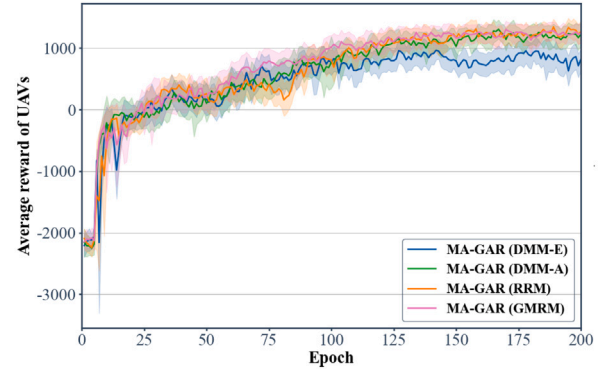


Fig. 10. Convergence of UAV agents' reward under different mobility models ($U = 4, M = 10$).

model (RRM), and the Gaussian-Markov random model (GMRM) proposed in Section 3.1.3. The results indicate that under both RRM and GMRM, MA-GAR converges to similar reward values, demonstrating that the algorithm can stably and efficiently optimize energy-efficient flight strategies even when MT's movement is highly random. Under DMM-A, MA-GAR still maintains a reward value similar to that of RRM and GMRM. In contrast, under DMM-E, the convergence reward value of MA-GAR is lower than that under DMM-A, RRM, and GMRM. This is because under DMM-E, each MT has a different initial movement direction, thereby reducing the probability that MTs form clusters (MT groups) as they move. Due to the UAV-BSS' limited coverage range, the DMM-E scenario with fewer MT groups makes it harder for UAV-BSSs to cover more MT groups. Ultimately, this results in lower convergence performance of MA-GAR in the DMM-E scenario.

Fig. 11 illustrates the trade-off between system throughput and UAV-BS energy consumption in MA-GAR, demonstrating the dynamic focusing effect of the weight coefficient α on the optimization objective. For ease of analysis, the figure labels the right vertical axis (energy consumption) as a negative value. As the weight coefficient α decreases, MA-GAR significantly reduces UAV energy consumption at the cost of system throughput. Conversely, when α increases, MA-GAR tends to prioritize throughput improvement at the expense of higher energy consumption. At $\alpha \approx 0.6$, an ideal equilibrium point appears on the trade-off curve, where MA-GAR achieves the best compromise between throughput and energy consumption. In the practical implementation of UAV-BS, the specific value of α can be tested and adjusted according to actual conditions to better adapt to different mission objectives.

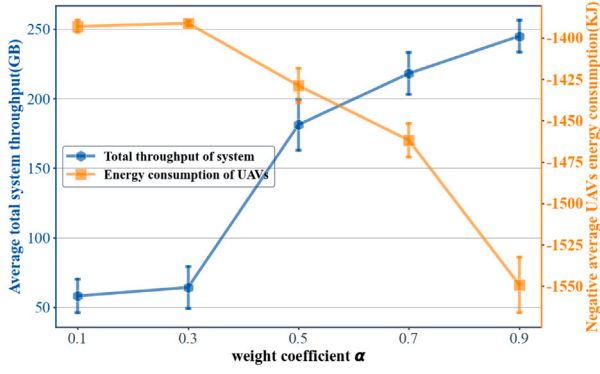


Fig. 11. System throughput and energy consumption vs the weight coefficient α .

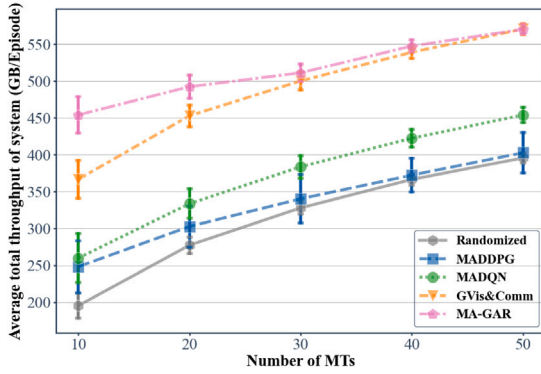


Fig. 12. Impact of the number of MTs on total system throughput with a fixed number of UAVs ($U = 10$).

5.3. Performance evaluation

Fig. 12 shows the total system throughput of different algorithms with 10 UAVs under varying numbers of MTs. As shown in the results, with the number of MTs increasing, the total system throughput gradually increases, and our proposed MA-GAR outperforms all benchmark algorithms, but our proposed MA-GAR algorithm consistently maintains the highest total system throughput across different numbers of MTs. Additionally, the performance growth rate of all algorithms generally shows a decreasing trend. This is because, as the number of MTs increases, more MTs enter the blind spots of UAV-BSs, and due to the irregular movement, it becomes difficult for UAV-BSs to provide service for them.

We next test the UAVs' energy consumption with 10 UAVs under varying numbers of MTs. As shown in Fig. 13, with the number of MTs increasing, the fluctuation of all energy consumption curves is not obvious; only randomized scheduling requires a large amount of energy for flight, so its energy consumption is larger. This indicates that all algorithms have learned more energy-effective trajectory strategies. During each trajectory planning, UAV-BSs consider the energy cost of decisions and their immediate and future benefits. As the number of MTs increases and distribution becomes denser, UAV-BSs are not inclined to expend much energy to search for other MTs. Hence, UAVs' energy consumption does not increase with the number of MTs. However, due to the efficient trajectory planning strategy, our proposed MA-GAR consistently maintains lower energy consumption than all benchmarks across different numbers of MTs.

Fig. 14 shows the total system throughput of different algorithms with 20 MTs under varying numbers of UAVs. From the results, we can see that the total system throughput increases with the number of

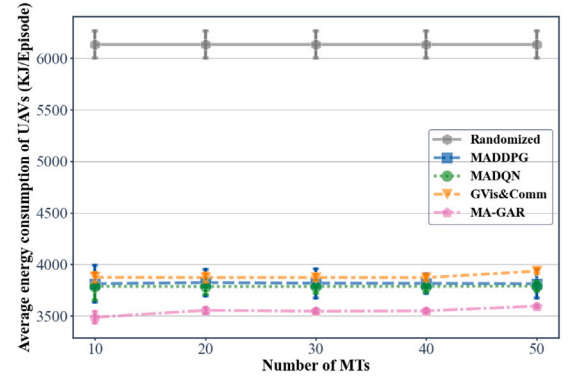


Fig. 13. Impact of the number of MTs on UAV energy consumption with a fixed number of UAVs ($U = 10$).

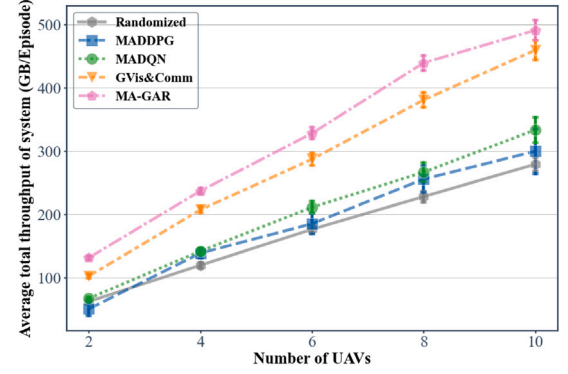


Fig. 14. Impact of the number of UAVs on total system throughput with a fixed number of MTs ($M = 20$).

UAVs. Our proposed MA-GAR algorithm achieves higher total system throughput than other benchmark algorithms across all numbers of UAVs. This is because, as the number of UAV-BSs increases, more MTs are covered, which increases total system throughput. However, our proposed MA-GAR produces better planning strategies and achieves better scheduling services regardless of the number of UAVs.

Fig. 15 shows the energy consumption of different algorithms with a fixed 20 MTs under varying numbers of UAVs. As shown in the results, the energy consumption of all algorithms increases as the number of UAVs grows. This is because deploying more UAVs is bound to cost more energy. However, our proposed MA-GAR consistently maintains the lowest energy consumption and exhibits the slowest growth rate across different numbers of UAVs. With the expansion of the UAV scale, MA-GAR can save more energy than other algorithms. It is more suitable for the deployment of large-scale UAVs.

5.4. Trajectory visualization

To explore the execution capability of UAV-BSs trained by MA-GAR, we tested their performance under two different environment scale settings. Figs. 16 and 17 show the visualization results of UAV-BSs' execution processes. These figures include UAV-BSs' trajectories, MTs' trajectories, the sensing and coverage areas of UAV-BSs at the final time slot, and the scheduling situation with MTs at the initial and final time slots over a service period.

Scenario I is an environment with 2 UAVs and 2 MTs. As shown in Fig. 16, UAV-1 initially covers an MT and successfully schedules it. Over time, the MT starts moving randomly. To prevent it from leaving its coverage area, UAV-1 continuously adjusts its trajectory to follow

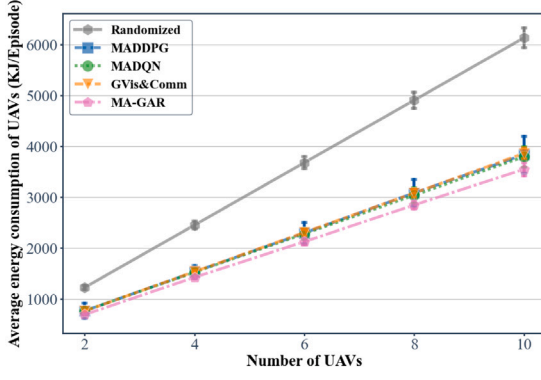


Fig. 15. Impact of the number of UAVs on their energy consumption ($M = 20$).

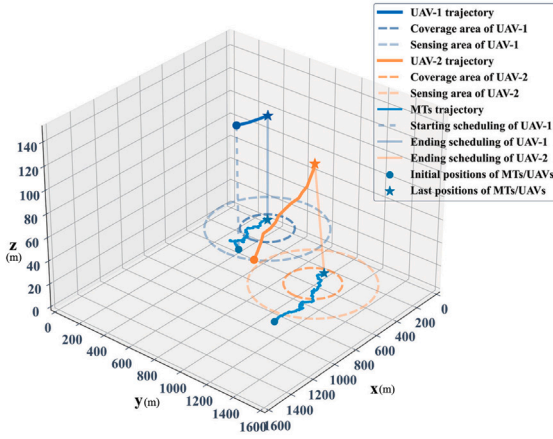


Fig. 16. Scenario I with two UAVs, two MTs and 4,000s duration.

the MT in real time, maintaining stable communication coverage. UAV-2 is placed in the edge of the area and does not sense any MTs at the beginning. Then UAV-2 first roams in the air, searching for MTs. Upon finding the nearest MT, UAV-1 quickly flies towards it, bringing it into coverage, and follows the MT in real-time to provide stable service like UAV-1.

Fig. 17 shows the execution process for Scenario II with 2 UAVs and 8 MTs. At the beginning, two groups (each has 4 MTs) are deployed in the area, UAV-1 is placed on the edge of the area and does not sense any MTs initially, UAV-2 covers one group of MTs. At the beginning of the service, UAV-1 first ascends to increase its coverage area, quickly searching for and covering a group of MTs. Upon finding the nearest group, UAV-1 descends and approaches the group, choosing to serve the three denser MTs, an MT in this group is out of the coverage area of UAV-1. Please note that lowering altitude allows the MTs to receive higher-quality wireless signals to achieve more efficient communication. When UAV-1 reaches the group served by UAV-2, UAV-2 relinquishes its current service to cooperate better and begins searching for other groups. Upon finding the right group, UAV-2, like UAV-1, follows the three denser MTs in real-time to provide the communication service.

5.5. Complexity analysis

Since MA-GAR adopts the actor-critic framework, this paper splits MA-GAR into actor and critic networks and analyzes their complexity separately. In addition, the complexity of the classical benchmark algorithms MADQN and MADDPG is also comparatively analyzed, as shown in Table 4, where U, M denotes the number of UAV-BSs and MTs,

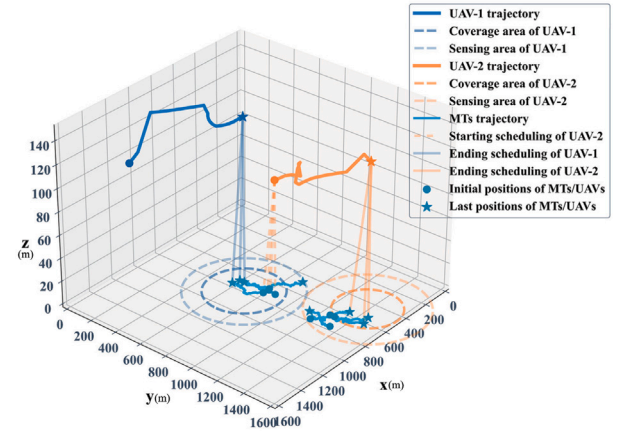


Fig. 17. Scenario II with two UAVs, eight MTs, group placement and 2,000s duration.

Table 4

Model complexity analysis.

Model	Complexity
MADQN	$O(U^2 D_{obs}^{uav} + U M D_{obs}^{mt})$
MADDPG's Actor	$O(U^2 D_{obs}^{uav} + U M D_{obs}^{mt})$
MADDPG's Critic	$O(U^2 (D_{obs}^{uav} + D_{act}) + U M D_{obs}^{mt})$
GVIs&Comm	$O(U U_{sns} D_{obs}^{uav} + U M_{sns} D_{obs}^{mt})$
MA-GAR's Actor	$O(U U_{sns} D_{obs}^{uav} + U M_{sns} D_{obs}^{mt})$
MA-GAR's Critic	$O(2U U_{sns} (D_{obs}^{uav} + D_{act}) + 2U M_{sns} D_{obs}^{mt})$

respectively, U_{sns}, M_{sns} denotes the number of other UAV-BSs and MTs sensed by the UAV agent, respectively, $D_{obs}^{uav}, D_{obs}^{mt}$ denotes the dimension size of the observation space of the UAV agent, and D_{act} denotes the dimension size of the action space. The observation and action spaces are related to the definitions in the POMDP. It can be observed that the MA-GAR proposed in this paper has lower complexity because the graph structure has good scalability and can be flexibly adapted to the information of any dimension according to the observation of the UAV agent.

Finally, using the same parameter settings as in Section 5.2, we also recorded the running times of MA-GAR and all benchmark algorithms. The average convergence time of MA-GAR was 586 ± 11 min, while the average convergence times for GVIs&Comm, MADDPG, and MA-DQN were 130 ± 15 min, 61 ± 16 min, and 35 ± 10 min, respectively. The higher running time of MA-GAR primarily stems from its more complex structural design, particularly the introduced multi-head graph attention mechanism and recurrent architecture. The repeated computation of attention values within these mechanisms, along with the processing of sequential samples, incurs additional computational overhead. We believe that, despite the increased time cost, this is a reasonable trade-off for MA-GAR to achieve superior performance.

5.6. Ablation analysis

To explore the impact of our designed recurrent multi-agent actor network (A-R) and graph feature encoding module with multi-head attention mechanism (AC-AGFEM), we conducted ablation experiments in an environment with 4 UAVs and 10 MTs. We analyze the impact of each module on MA-GAR, i.e., without A-R, without AC-AGFEM, and without both of them. The results are shown in Fig. 18.

MA-GAR without A-R exhibits more fluctuations throughout the training process, with a larger curve shadow and multiple reward drops in the mid-to-late stages. It converges at about 130 epochs. In contrast, MA-GAR converges at about 100 epochs, with a higher converged reward value than MA-GAR without A-R. It also performs more stably throughout the training process, with smaller curve shadows and no

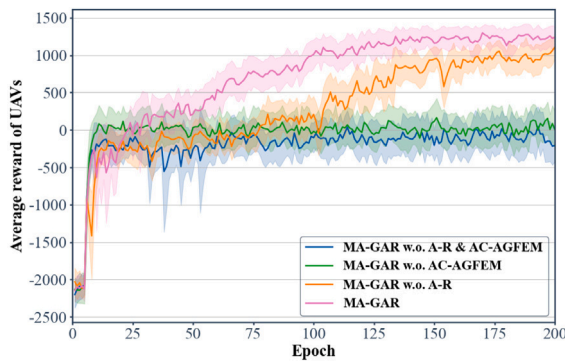


Fig. 18. Convergence of reward for UAV agents in MA-GAR with different modules removed ($U = 4$, $M = 10$).

reward drops in the mid-to-late stages. This is because the A-R component of MA-GAR allows UAV agents to remember historical interaction information. This enables them to infer global information from their partially observed environment, take more accurate actions, and thus achieve more stable performance.

MA-GAR without AC-AGFEM is relatively stable throughout the training process and converges quickly, but it has a lower reward throughout the training process. In contrast, the reward of MA-GAR is much higher. This is because MA-GAR without AC-AGFEM uses traditional vector representations for UAV agent observations and deep neural networks (DNNs) for learning. The proposed heterogeneous graph representation of observations can express more information than traditional vector representation, such as potential spatial relationships between UAVs and network entities. The strong topology description ability of heterogeneous graphs can flexibly handle observations of different dimensions, and GNNs have advantages over DNNs in dynamic, stochastic, and partially observable environments.

Evidently, MA-GAR without A-R and AC-AGFEM not only exhibits severe fluctuations and reward drops but also achieves the lowest reward among all algorithms. This proves that our designed A-R and AC-AGFEM are essential for MA-GAR, which is crucial in handling extremely stochastic, complex, dynamic, and partially observable environments.

6. Conclusion

In this paper, we have explored the energy-efficient communication problem in the UAV-assisted air-ground integrated network. The graph-based information representation method and attention mechanism designed in our MA-GAR can flexibly and efficiently learn the characteristics with fluctuating dimensions in POMDP. Furthermore, the introduced CTDE mechanism can avoid additional energy consumption of UAVs caused by direct training. The simulation experiments were conducted, and the results demonstrated that our proposed MA-GAR can reduce energy consumption by up to about 6.1% compared to benchmark algorithms while ensuring high throughput and fairness. In this paper, the lack of network parameter-sharing capability among UAV agents limits their cooperative efficiency. Therefore, future research will introduce this mechanism, along with the dynamic adjustment mechanism for the optimization objective weight, to explore a more energy-efficient and effective cooperative trajectory planning strategy.

CRedit authorship contribution statement

Cong Wang: Writing – review & editing, Validation. **Menglong Dong:** Writing – original draft. **Ying Yuan:** Visualization. **Guorui Li:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] L. Qiao, Y. Li, D. Chen, S. Serikawa, M. Guizani, Z. Lv, A survey on 5G/6G, AI, and robotics, *Comput. Electr. Eng.* 95 (2021) 107372, <http://dx.doi.org/10.1016/j.compeleceng.2021.107372>.
- [2] X. Jiang, M. Sheng, N. Zhao, C. Xing, W. Lu, X. Wang, Green UAV communications for 6G: A survey, *Chin. J. Aeronaut.* 35 (2022) 19–34, <http://dx.doi.org/10.1016/j.cja.2021.04.025>.
- [3] Q. Shen, J. Peng, W. Xu, Y. Sun, W. Liang, L. Chen, Q. Zhao, X. Jia, Fair communications in UAV networks for rescue applications, *IEEE Internet Things J.* 10 (2023) 21013–21025, <http://dx.doi.org/10.1109/JIOT.2023.3284813>.
- [4] M. Li, X. Liu, H. Wang, Completion time minimization considering GNs' energy for UAV-assisted data collection, *IEEE Wirel. Commun. Lett.* 12 (2023) 2128–2132, <http://dx.doi.org/10.1109/LWC.2023.3309625>.
- [5] C. Wang, X. Chai, S. Peng, Y. Yuan, G. Li, Deep reinforcement learning with entropy and attention mechanism for D2D-assisted task offloading in edge computing, *IEEE Trans. Serv. Comput.* (2024) 1–13, <http://dx.doi.org/10.1109/TSC.2024.3495503>.
- [6] S. Ahmad, J. Zhang, A. Nauman, A. Khan, K. Abbas, B. Hayat, Deep-EERA: DRL-based energy-efficient resource allocation in UAV-empowered beyond 5G networks, *Tsinghua Sci. Technol.* 30 (2025) 418–432, <http://dx.doi.org/10.26599/TST.2024.9010071>.
- [7] X. Tang, F. Chen, F. Wang, Z. Jia, Disaster-resilient emergency communication with intelligent air-ground cooperation, *IEEE Internet Things J.* 11 (2024) 5331–5346, <http://dx.doi.org/10.1109/JIOT.2023.3305549>.
- [8] Q. Zhang, Y. Luo, H. Jiang, K. Zhang, Aerial edge computing: A survey, *IEEE Internet Things J.* 10 (2023) 14357–14374, <http://dx.doi.org/10.1109/JIOT.2023.3263360>.
- [9] N. Cheng, S. Wu, X. Wang, Z. Yin, C. Li, W. Chen, F. Chen, AI for UAV-assisted IoT applications: A comprehensive review, *IEEE Internet Things J.* 10 (2023) 14438–14461, <http://dx.doi.org/10.1109/JIOT.2023.3268316>.
- [10] G. Wu, Z. Xu, H. Zhang, S. Shen, S. Yu, Multi-agent DRL for joint completion delay and energy consumption with queuing theory in MEC-based IIoT, *J. Parallel Distrib. Comput.* 176 (2023) 80–94, <http://dx.doi.org/10.1016/j.jpdc.2023.02.008>.
- [11] X. Yan, X. Fang, C. Deng, X. Wang, Joint optimization of resource allocation and trajectory control for mobile group users in fixed-wing UAV-enabled wireless network, *IEEE Trans. Wirel. Commun.* 23 (2024) 1608–1621, <http://dx.doi.org/10.1109/TWC.2023.3290748>.
- [12] M. Kishk, A. Bader, M.-S. Alouini, Aerial base station deployment in 6G cellular networks using tethered drones: The mobility and endurance tradeoff, *IEEE Veh. Technol. Mag.* 15 (2020) 103–111, <http://dx.doi.org/10.1109/MVT.2020.3017885>.
- [13] Q. Zeng, Y. Jia, C. Li, L. Liu, 3-D deployment of UAV-BSs for effective communication coverage, *IEEE Internet Things J.* 11 (2024) 25162–25172, <http://dx.doi.org/10.1109/JIOT.2024.3392950>.
- [14] N. Nouri, F. Fazel, J. Abouei, K.N. Plataniotis, Multi-UAV placement and user association in uplink MIMO ultra-dense wireless networks, *IEEE Trans. Mob. Comput.* 22 (2023) 1615–1632, <http://dx.doi.org/10.1109/TMC.2021.3108960>.
- [15] S. Wang, S. Zhao, H. Gui, X. He, Z. Lu, B. Chen, Z. Fan, S. Pang, Energy-efficient collaborative task offloading in multi-access edge computing based on deep reinforcement learning, *Ad Hoc Networks* 169 (2025) 103743, <http://dx.doi.org/10.1016/j.adhoc.2024.103743>, URL <https://www.sciencedirect.com/science/article/pii/S1570870524003548>.
- [16] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, P.S. Yu, A comprehensive survey on graph neural networks, *IEEE Trans. Neural Networks Learn. Syst.* 32 (2021) 4–24, <http://dx.doi.org/10.1109/TNNLS.2020.2978386>.

- [17] Y. Wang, Y. Li, J. Lei, F. Shang, Robust and energy-efficient RPL optimization algorithm with scalable deep reinforcement learning for IIoT, *Comput. Netw.* 255 (2024) 110894, <http://dx.doi.org/10.1016/j.comnet.2024.110894>, URL <https://www.sciencedirect.com/science/article/pii/S1389128624007266>.
- [18] Y. Song, Q. Zeng, G. Chen, G. Yuan, H. Duan, Multi-agent proximal policy optimization based efficient user association and resource allocation in UAV-assisted heterogeneous cellular networks, *Comput. Commun.* 238 (2025) 108172, <http://dx.doi.org/10.1016/j.comcom.2025.108172>, URL <https://www.sciencedirect.com/science/article/pii/S014036642500129X>.
- [19] F. Tang, X. Chen, T.K. Rodrigues, M. Zhao, N. Kato, Survey on digital twin edge networks (DITEN) toward 6G, *IEEE Open J. Commun. Soc.* 3 (2022) 1360–1381, <http://dx.doi.org/10.1109/OJCOMS.2022.3197811>.
- [20] S. Mihai, M. Yaqoob, D.V. Hung, W. Davis, P. Towakel, M. Raza, M. Karamanoglu, B. Barn, D. Shetve, R.V. Prasad, H. Venkataraman, R. Trestian, H.X. Nguyen, Digital twins: A survey on enabling technologies, challenges, trends and future prospects, *IEEE Commun. Surv. Tutor.* 24 (2022) 2255–2291, <http://dx.doi.org/10.1109/COMST.2022.3208773>.
- [21] Q. Wu, R. Zhang, Common throughput maximization in UAV-enabled OFDMA systems with delay consideration, *IEEE Trans. Commun.* 66 (2018) 6614–6627, <http://dx.doi.org/10.1109/TCOMM.2018.2865922>.
- [22] J. Wang, H. Zhang, X. Zhou, W. Liu, D. Yuan, Joint resource allocation and trajectory design for energy-efficient UAV assisted networks with user fairness guarantee, *IEEE Internet Things J.* 11 (2024) 23835–23849, <http://dx.doi.org/10.1109/JIOT.2024.3386861>.
- [23] N. Zhao, Y.-C. Liang, D. Niyato, Y. Pei, M. Wu, Y. Jiang, Deep reinforcement learning for user association and resource allocation in heterogeneous cellular networks, *IEEE Trans. Wirel. Commun.* 18 (2019) 5141–5152, <http://dx.doi.org/10.1109/TWC.2019.2933417>.
- [24] Z.A. Arain, X. Qiu, C. Xu, M. Wang, M. Abdul Rahim, Energy-aware MPTCP scheduling in heterogeneous wireless networks using multi-agent deep reinforcement learning techniques, *Electronics* 12 (2023) <http://dx.doi.org/10.3390/electronics12214496>, URL <https://www.mdpi.com/2079-9292/12/21/4496>.
- [25] X. Song, M. Cheng, L. Lei, Y. Yang, Multitask and multiobjective joint resource optimization for UAV-assisted air-ground integrated networks under emergency scenarios, *IEEE Internet Things J.* 10 (2023) 20342–20357, <http://dx.doi.org/10.1109/JIOT.2023.3284425>.
- [26] X. Zhang, H. Zhao, J. Wei, C. Yan, J. Xiong, X. Liu, Cooperative trajectory design of multiple UAV base stations with heterogeneous graph neural networks, *IEEE Trans. Wirel. Commun.* 22 (2023) 1495–1509, <http://dx.doi.org/10.1109/TWC.2022.3204794>.
- [27] B. Li, W. Liu, W. Xie, N. Zhang, Y. Zhang, Adaptive digital twin for UAV-assisted integrated sensing, communication, and computation networks, *IEEE Trans. Green Commun. Netw.* 7 (2023) 1996–2009, <http://dx.doi.org/10.1109/TGCN.2023.3298039>.
- [28] W. Liu, B. Li, W. Xie, Y. Dai, Z. Fei, Energy efficient computation offloading in aerial edge networks with multi-agent cooperation, *IEEE Trans. Wirel. Commun.* 22 (2023) 5725–5739, <http://dx.doi.org/10.1109/TWC.2023.3235997>.
- [29] B. Wang, Y. Sun, H. Jung, L.D. Nguyen, N.-S. Vo, T.Q. Duong, Digital twin-enabled computation offloading in UAV-assisted MEC emergency networks, *IEEE Wirel. Commun. Lett.* 12 (2023) 1588–1592, <http://dx.doi.org/10.1109/LWC.2023.3283912>.
- [30] A. Tanaka, B. Du, R. Li, Digital twin-empowered post-disaster movable base station deployment optimization, *IEEE Commun. Mag.* (2025) 1–7, <http://dx.doi.org/10.1109/MCOM.001.2500070>.
- [31] W. Xie, H. Yang, Z. Xiong, Resource allocation for UAV-assisted anti-jamming semantic D2D networks: A graph reinforcement learning approach, *Comput. Netw.* 269 (2025) 111463, <http://dx.doi.org/10.1016/j.comnet.2025.111463>, URL <https://www.sciencedirect.com/science/article/pii/S138912862500430X>.
- [32] Z. Sun, Y. Mo, C. Yu, Graph-reinforcement-learning-based task offloading for multiaccess edge computing, *IEEE Internet Things J.* 10 (2023) 3138–3150, <http://dx.doi.org/10.1109/JIOT.2021.3123822>.
- [33] Q. Liu, L. Shi, L. Sun, J. Li, M. Ding, F. Shu, Path planning for UAV-mounted mobile edge computing with deep reinforcement learning, *IEEE Trans. Veh. Technol.* 69 (2020) 5723–5728, <http://dx.doi.org/10.1109/TVT.2020.2982508>.
- [34] L. Wang, K. Wang, C. Pan, W. Xu, N. Aslam, L. Hanzo, Multi-agent deep reinforcement learning-based trajectory planning for multi-UAV assisted mobile edge computing, *IEEE Trans. Cogn. Commun. Netw.* 7 (2021) 73–84, <http://dx.doi.org/10.1109/TCCN.2020.3027695>.
- [35] S. Yin, F.R. Yu, Resource allocation and trajectory design in UAV-aided cellular networks based on multiagent reinforcement learning, *IEEE Internet Things J.* 9 (2022) 2933–2943, <http://dx.doi.org/10.1109/JIOT.2021.3094651>.
- [36] Z. Yao, S. Xia, Y. Li, G. Wu, Cooperative task offloading and service caching for digital twin edge networks: A graph attention multi-agent reinforcement learning approach, *IEEE J. Sel. Areas Commun.* 41 (2023) 3401–3413, <http://dx.doi.org/10.1109/JSAC.2023.3310080>.
- [37] R.J. Kerczewski, J.H. Griner, Control and non-payload communications links for integrated unmanned aircraft operations, in: 2012 Joint Conference-18th Ka and Broadband, 2012.
- [38] A. Al-Hourani, S. Kandeepan, S. Lardner, Optimal LAP altitude for maximum coverage, *IEEE Wirel. Commun. Lett.* 3 (2014) 569–572, <http://dx.doi.org/10.1109/LWC.2014.2342736>.
- [39] A. Fotouhi, M. Ding, M. Hassan, Understanding autonomous drone maneuverability for Internet of Things applications, in: 2017 IEEE 18th International Symposium on a World of Wireless, Mobile and Multimedia Networks, WoWMoM, 2017, pp. 1–6, <http://dx.doi.org/10.1109/WoWMoM.2017.7974336>.
- [40] Y. Zeng, J. Xu, R. Zhang, Energy minimization for wireless communication with rotary-wing UAV, *IEEE Trans. Wirel. Commun.* 18 (2019) 2329–2345, <http://dx.doi.org/10.1109/TWC.2019.2902559>.
- [41] S. Brody, U. Alon, E. Yahav, How attentive are graph attention networks? in: *International Conference on Learning Representations*, 2022.
- [42] M.L. Betalo, S. Leng, H.N. Abishu, A.M. Seid, M. Fakirah, A. Erbad, M. Guizani, Multi-agent DRL-based energy harvesting for freshness of data in UAV-assisted wireless sensor networks, *IEEE Trans. Netw. Serv. Manag.* (2024) 1, <http://dx.doi.org/10.1109/TNSM.2024.3454217>.
- [43] Z. Ye, K. Wang, Y. Chen, X. Jiang, G. Song, Multi-UAV navigation for partially observable communication coverage by graph reinforcement learning, *IEEE Trans. Mob. Comput.* 22 (2023) 4056–4069, <http://dx.doi.org/10.1109/TMC.2022.3146881>.